REPUBLIC POLYTECHNIC

Lesson 21
Demand Forecasting: MA & Forecast Error

E211 – Operations Planning II

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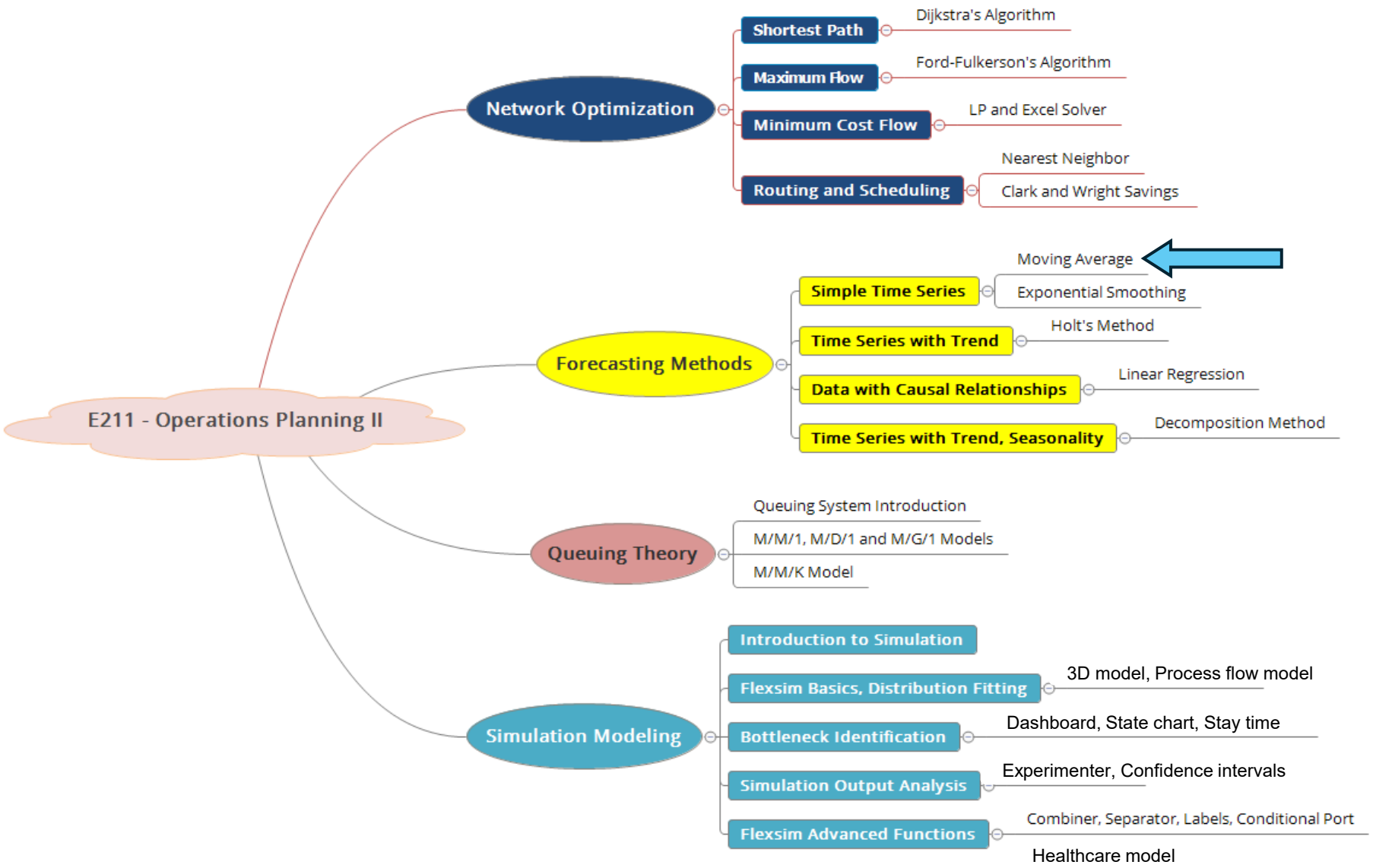
E211 Topic Tree

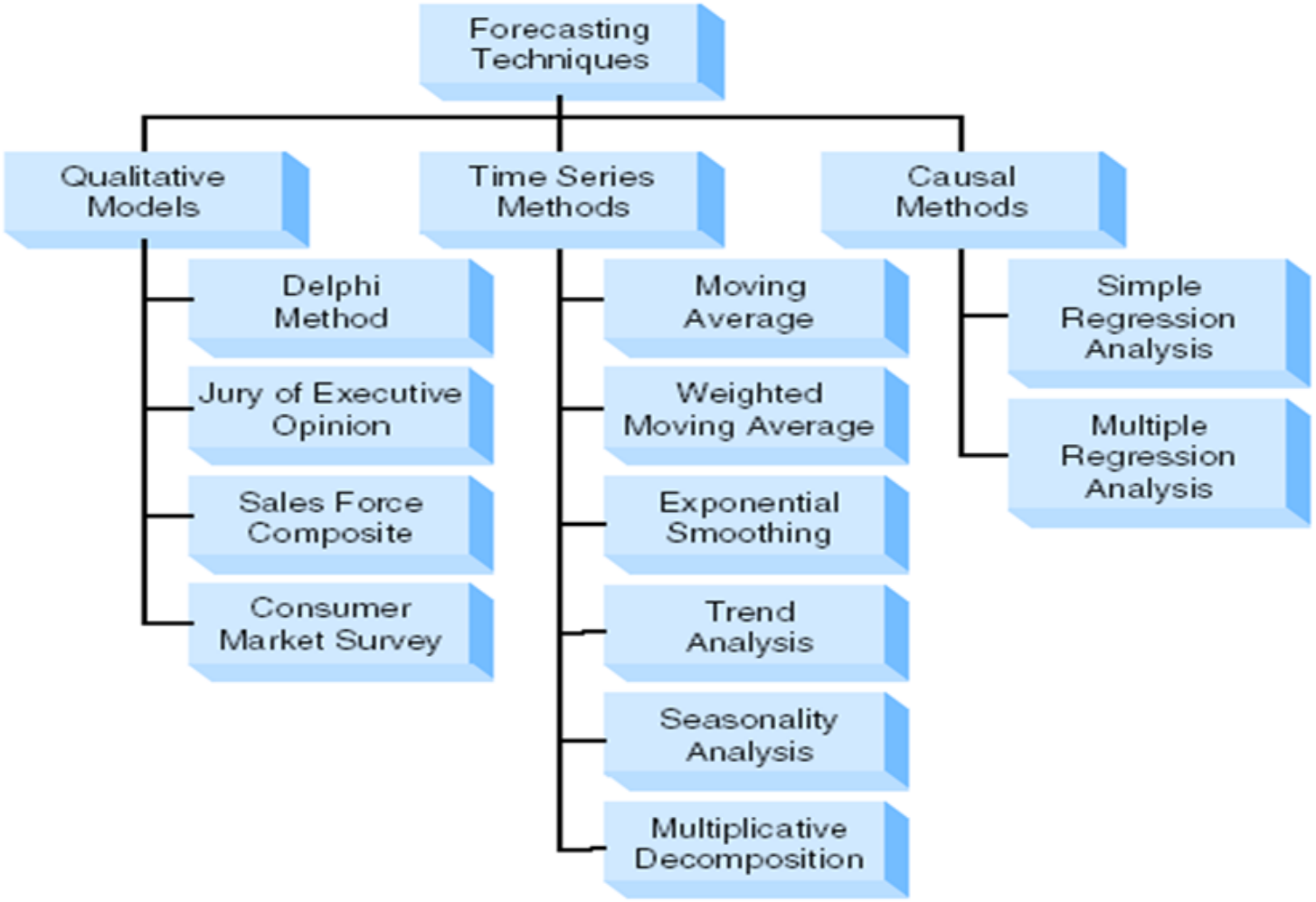


Introduction to Forecasting

Importance of Forecasting

- Companies need to forecast product demand, resource availability, technological changes and business cycles
- Forecasting is important for all strategic and operations planning decisions in a supply chain
  - Forecasts of product demand, materials, labour, financing are important inputs to scheduling, acquiring resources, and determining resource requirements.
  - For example: if a company carries more inventory than needed, they're wasting resources; while if it carries less than needed, they're not able to satisfy the customer needs.

Types of Forecasts



Forecasting Methods

Qualitative Forecast

- Subjective in nature
- Expert judgment
  - Consensus Panel
  - Delphi Method
- Grassroots Forecasting
- Market Research Survey

Quantitative Forecast

- Mathematical or simulation models
- Causal forecasting
- Time series forecasting

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Qualitative Forecasting

Expert Judgment

- Consensus Panel (focus group)
  - Bring experts together for discussion until consensus emerges
  - Problems with group dynamics may override forecast accuracy

- Delphi Method
  - Coordinator requests/receives forecasts from experts (through emails, surveys)
  - Coordinator determines median response and the range of middle 50% of answers
  - Coordinator requests explanations from experts outside middle 50%
  - Median response, middle 50% range, and explanations sent to experts
  - Process repeated till consensus reached (time consuming)

Qualitative Forecasting

Grassroots/ Salesforce Forecasting

- Bottom-up approach where the opinions and estimates of the sales force are solicited
- Advantage
  - Provides a lot of detailed knowledge such as insights into regional trends and individual customer needs
- Disadvantages
  - Conflict of Interest
  - Accuracy depends on quality and training of sales staff

Market Research Survey

- Gather direct feedback from current and potential customers to estimate future demand
- Advantage
  - Well-designed survey representative of consumer opinions
- Disadvantage
  - May be costly and time consuming

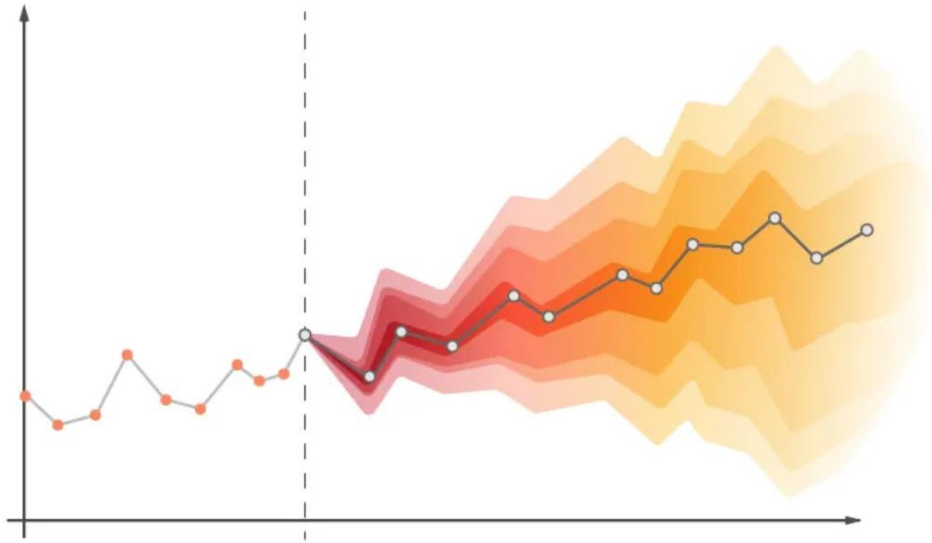
Quantitative Forecasting

- Uses numerical values to generate the forecast
- Two types: **Causal forecasting** and **Time-series forecasting**
- Can be systematically modified and requires a large quantity of data
- Frequent updates are required
- Validation is difficult (how good/accurate is the model?)

Time Series Forecasting

Time Series Forecasting

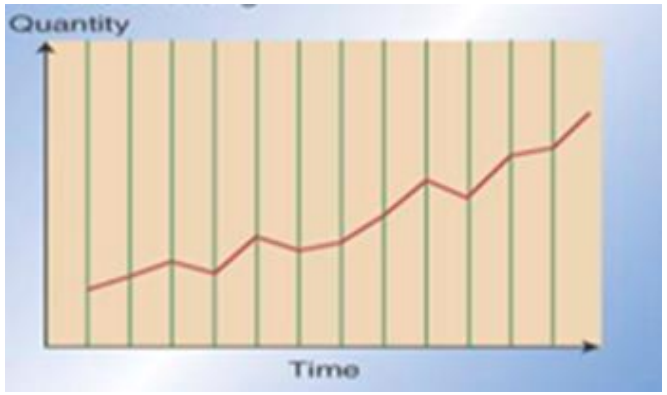
- Predicting the future from the historical data
- Based on a sequence of evenly spaced data points gathered sequentially in time (weekly, monthly, quarterly, etc.) known as time series
- Causal variables and qualitative factors are not considered



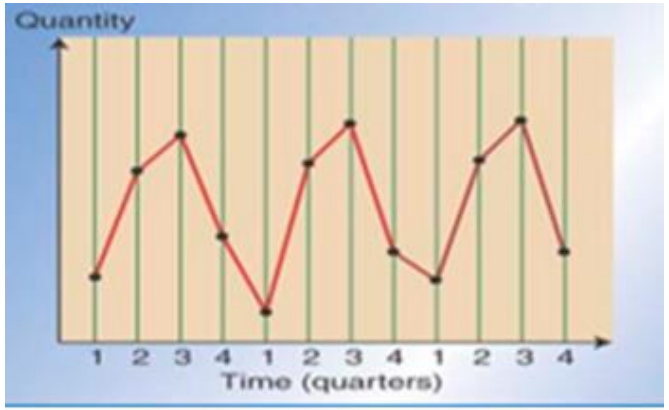
"Forecasting is the art of saying what will happen and then explaining why it didn't!"

Components of a Time Series

- **Trend** – long term upward or downward movement of data over time

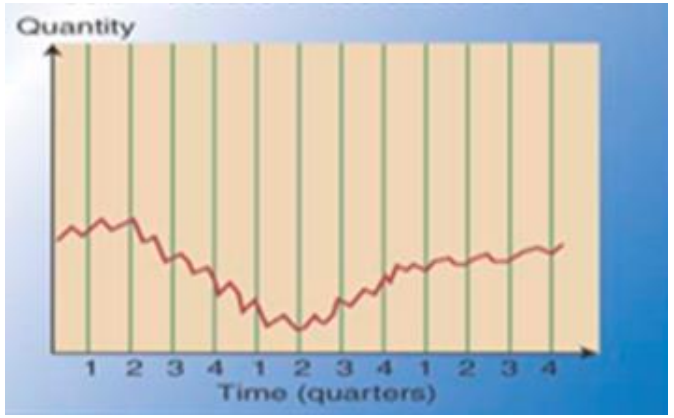


- **Seasonality** – pattern that repeats itself at fixed intervals such as every one year (e.g. sales of winter clothing)

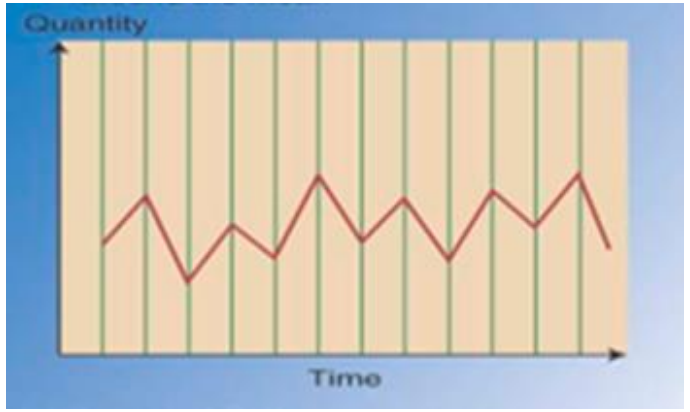


Components of a Time Series

- **Cyclicity** – fluctuations that occur over longer periods of time. Such as unpredictable pattern due to economic or business climate (usually every few years)

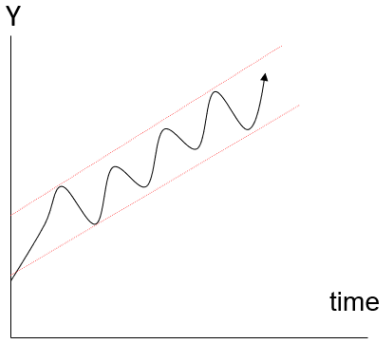


- **Random/ Irregular** – unpredictable or random variations, due to chance or unusual situations with no discernible patterns



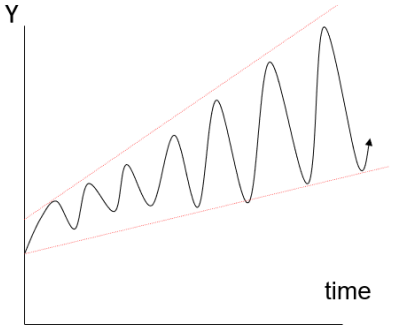
General Models

- Additive model: suitable for time series with constant seasonal variation
  - $Y_t = T_t + C_t + S_t + I_t$



Additive seasonality

- Multiplicative model: suitable for time series with increasing/decreasing seasonal variation
  - $Y_t = T_t * C_t * S_t * I_t$



Multiplicative seasonality

Where:

Y_t: The observed value of the time series at time t
T_t: The trend component in time t
C_t: The cyclical component in time t.
S_t: The seasonal component in time t.
I_t : The irregular component in time t.

Time Series Forecasting Methods

1. Moving Average (MA)
2. Weighted Moving Average (WMA)
3. Simple Exponential Smoothing (SES)
4. Trend-adjusted exponential smoothing (Holt's method)
5. Decomposition method
    1. Multiplicative decomposition method
    2. Additive decomposition method

Moving Average

Moving Average (MA)

- A simple method that provides quick forecast based on average of most recent N sets of data
- Each point is given equal weight so that short-term fluctuations (noises) are filtered out
- Formula for MA:

$$F_t = \frac{A_{t-N} + A_{t-N+1} + \ldots.. + A_{t-1}}{N}$$

Where:

F_t is the forecast in period t

N is the number of most recent periods used in the forecast.

A_{t-N} = previous N period's actual demand

Moving Average (MA)

- Uses the *N* most recent data points to predict
- May require extensive records of past data
- Will lag (*N*+1)/2 periods if trend is present
- What '*N*' should be chosen?
  - ➢**Small *N value*** is very responsive to the last observed demand point but choppier
  - ➢**Large *N value*** makes level less responsive (lags more) but smoother
- Is used to identify medium and long-term trends
  - E.g. stock traders used MA to determine support and resistance levels

Moving Average (MA) Exercise

The following table lists Company ABC's monthly sales of Product A for the past 12 months:

Month	Sales Figure
Oct-20	501
Nov-20	429
Dec-20	352
Jan-21	472
Feb-21	342
Mar-21	367
Apr-21	370
May-21	437
Jun-21	412
Jul-21	336
Aug-21	388
Sep-21	344

Using number of periods N=3, the moving average (N=3) will start from Jan-21.

Forecast (Jan-21) = (501+429+352)/3 = 427.33

Work out the forecasts from Feb-21 to Oct-21. Show workings for Oct-21 calculations.

Oct-21 = (336+388+344)/3 = 356

Plot the forecasted values with the actual sales figures and comment on the graphs.

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Moving Average (MA) Exercise

The following table lists Company ABC's monthly sales of Product A for the past 12 months:

Month	Sales Figure
Oct-20	501
Nov-20	429
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Sep-21	344

Using number of periods N=6, the moving average (N=6) will start from Jan-21.

Forecast (Apr-21) = (501+429+352+472+342+367)/6 = 410.5

Work out the forecasts from May-21 to Oct-21. Show workings for Oct-21 calculations.
Oct-21 = (370+437+412+336+388+344)/6 = 381.17

Add the forecasted values to the previous graphs (actual sales figures and MA, N=3) and comment on the graphs.
As N increases, the forecast becomes less responsive to changes in actual data and the graph becomes smoother.

Moving Average (MA) Exercise

Month	Sales Figure
Oct-20	501
Nov-20	429
Dec-20	352
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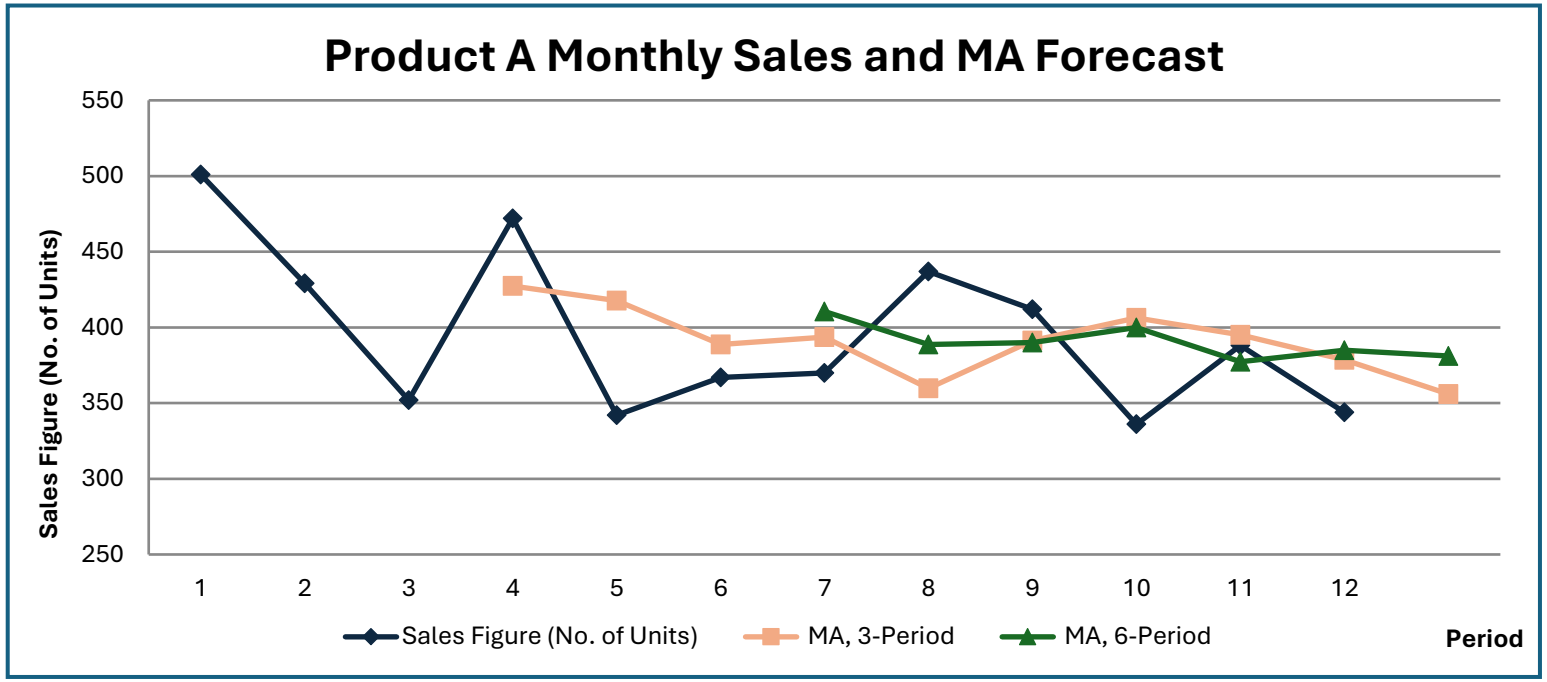

1. Forecast for Oct-21 is 356 (N=3) and 381.17 (N=6) respectively.

2. N=3 MA graph is more responsive to changes in sales figures compared with N=6 MA graph.

3. As N increases, the forecast becomes less responsive to changes in actual data and the graph becomes smoother.

Forecast Error

- Forecast will always deviate from the actual demand. Forecast Error is the difference between the predicted value and the actual value.
- A small error means more accurate forecast whereas a larger error suggests that the forecast model may need revision.
- Common forecast error measures:

Mean absolute deviation (MAD)	$MAD = \frac{1}{n}\sum_{t=1}^{n} \lvert A_t - F_t \rvert$
Mean absolute percent error (MAPE)	$MAPE = \frac{1}{n}\sum_{t=1}^{n} \frac{\lvert A_t - F_t \rvert}{A_t} \times 100\%$

MAD = |Actual Sales Figure – Forecast Value|

MAPE = |Actual Sales Figure – Forecast Value|/Actual Sales Figure * 100%

Where:
n is the total number of periods for which both actual and forecasted values exist.
A_t is the actual value at period t
F_t is the forecasted value at period t

Forecast Error Exercise

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Oct-20	501
Nov-20	429
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Using the previous set of sales data, calculate based on MA (N=6) forecast values the:

Forecast (Apr-21) = (501+429+352+472+342+367)/6 = 410.5

Absolute deviation in Apr-21

MAD = |Actual Sales Figure – Forecast Value|

= |370 – 410.5| = 40.5

Absolute percent error in Apr-21

MAPE = |Actual Sales Figure – Forecast Value|/Actual Sales Figure * 100%

= |370 – 410.5|/370 * 100%

= 10.95%

Forecast Error Exercise

Month	Sales Figure
Oct-20	501
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Using the previous set of sales data, calculate based on MA (N=6) forecast values the:

1. Absolute deviation in Apr-21
2. Absolute percent error in Apr-21

MA (N=6) forecast for Apr-21 = 410.5

1. Absolute deviation = | 370 – 410.5 | = 40.5
2. Absolute percent error = |370 – 410.5|/ 370 x 100% = 10.95%

Work out the MAD & MAPE for MA (N=3) and MA (N=6). Which MA is more accurate?

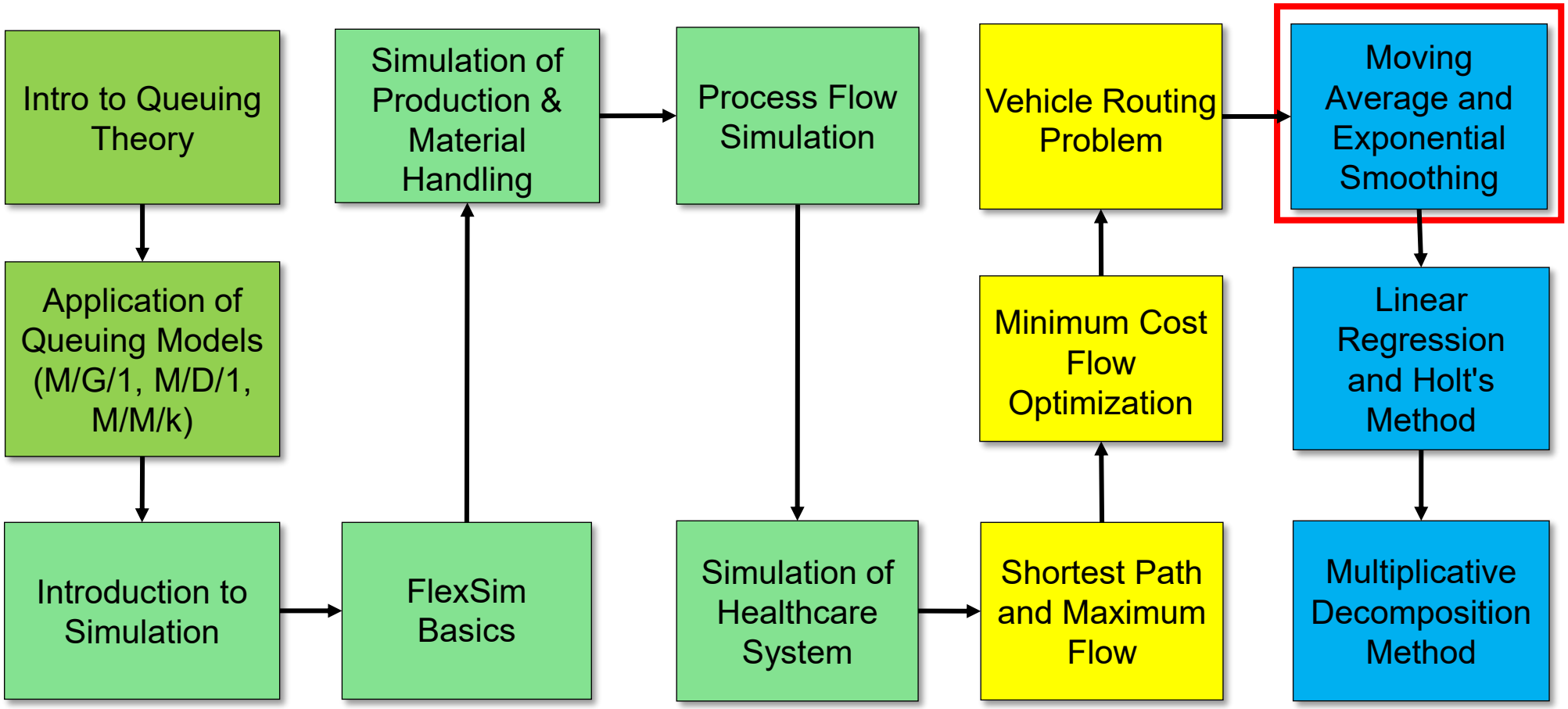
Forecast Error Exercise

MAD (MA 3-Period	MAD (MA 6-Period)	MAPE (MA 3-Period)	MAPE (MA 6-Period)
41.74	37.75	11.05%	10.18%

- MAD and MAPE calculations show that MA 6-Period forecasts have a lower average error compared with MA 3-Period forecasts.
- In this case, MA 6-Period forecasts provide a (slightly) better accuracy.
- Take note that MAPE calculates the average error as a percentage of actual values. This can be intuitive when it comes to error interpretation (compared with MAD) but may not be suitable for items with low demands.

Learning Outcomes

At the end of the lesson, students should be able to:

- Explain forecasting and time series analysis
- Describe the difference between qualitative and quantitative forecasting methods.
- Apply Moving Average to forecast a time series.
- Apply mean absolute deviation and mean absolute percent deviation to determine forecast errors.

E211 Progression